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Group 11: Spam Detection System (ML/NLP)

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Spam Detection System - Project Report

1. Problem Definition

1.1 Problem Statement

In today's digital age, unsolicited and malicious messages (spam) pose a significant threat to mobile phone users and email recipients worldwide. Spam messages waste user time, consume network bandwidth, and increasingly serve as vectors for phishing attacks, financial fraud, and malware distribution. Traditional rule-based spam filters struggle to keep pace with evolving spam techniques that use obfuscation, social engineering, and contextual evasion.

The problem: Develop an intelligent system capable of accurately distinguishing between legitimate (HAM) and spam messages using both statistical machine learning and cybersecurity rule-based analysis.

1.2 Real-World Importance

Impact Area	Description
Security	Prevents phishing attacks that steal personal and financial information
Productivity	Saves users hours wasted reading and deleting spam
Network Efficiency	Reduces unnecessary bandwidth consumption
Trust	Maintains user confidence in communication platforms
Financial Protection	Prevents SMS-based financial scams targeting vulnerable populations

1.3 Project Objectives

1. Develop a hybrid spam classification system combining ML and expert rule-based analysis
2. Achieve high accuracy (>95%) on the SMS Spam Collection dataset
3. Create an interactive web interface for real-time predictions
4. Provide batch processing capability for bulk message analysis
5. Deliver interpretable results explaining why a message is classified as spam

2. Dataset

2.1 Data Source

Dataset Name: SMS Spam Collection v.1

Source: **UCI Machine Learning Repository**

2.2 Dataset Statistics

Property	Value
Total Samples	5,574 messages
Spam Messages	747 (13.4%)
Ham Messages	4,827 (86.6%)
Format	CSV with two columns: label (spam/ham), message
Language	English (primarily)

2.3 Data Preprocessing

```python
def normalize_message(text):
    normalized = re.sub(r'\s+', ' ', str(text).strip().lower())
    return normalized
```

Preprocessing steps applied:

1. **Lowercase conversion** - Standardizes text case
2. **Whitespace normalization** - Removes extra spaces and line breaks
3. **Special character handling** - Preserves punctuation for spam indicators (e.g., multiple exclamation marks)
4. **No stop word removal** - Preserves context for spam detection

2.4 Train-Test Split

Split	Percentage	Number of Samples
Training	80%	4,459
Testing	20%	1,115

Method: Stratified random split to maintain class distribution

3. Modeling Approach

3.1 Baseline Model: Logistic Regression with TF-IDF

Why Logistic Regression as Baseline:

- Simple, interpretable, and fast to train
- Performs well on high-dimensional sparse text data
- Provides probability estimates for confidence scoring
- Serves as an effective benchmark for more complex models

Feature Engineering - TF-IDF Vectorization:

Parameter	Value
max_features	5,000
diangram_range	(1, 2)
analyzer	word
stop words	English

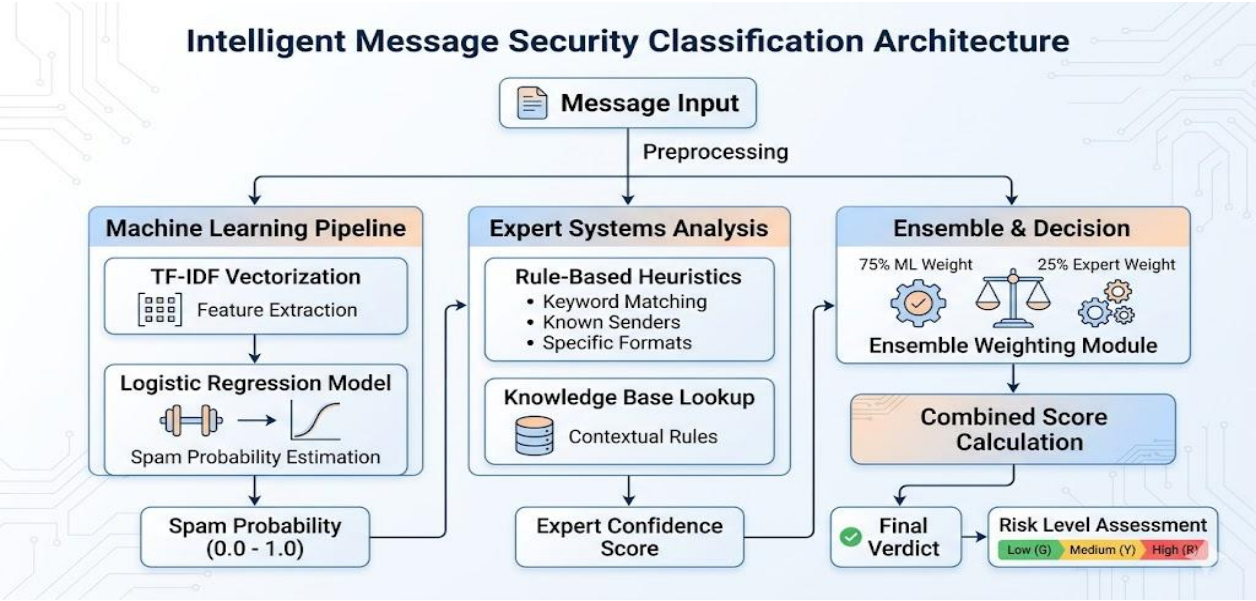
TF-IDF Formula:

TF-IDF(t,d) = TF(t,d) × log(N / DF(t))

Where:

- TF(t,d) = Term frequency in document d
- DF(t) = Number of documents containing term t
- N = Total number of documents

3.2 Advanced Model: Hybrid System (ML + Expert Rules)



Expert Rule Categories:

Category	Indicators Detected	Weight
Urgency	"act now", "immediate", "only X hours left"	+0.15 per match
Obfuscation	"w*i*n*n*e*r", "*f*r*e*e", "p*r*i*z*e"	+0.20 per match
Suspicious URLs	bit.ly, tiny url, t.co, short links	+0.30
Phone Numbers	International/scam formats	+0.15 per match
Money/Prizes	"win", "cash", "prize", "free entry"	+0.25 per match
Phishing Keywords	"verify", "account", "login", "bank"	+0.15 per match
Poor Grammar	Excessive !!!, ALL CAPS words	+0.10
Personal Tone	"hey", "thanks", "see you"	-0.20
Sender Identity	"mom", "friend", named references	-0.15

3.3 Model Justification

Model	Strengths	Weaknesses	Why Chosen
Logistic Regression	Interpretable, fast, probabilistic	Linear decision boundary	Baseline performance
Hybrid System	Combines statistical + semantic analysis	More complex to tune	Leverages both approaches

Justification for Hybrid Approach:

- ML captures statistical patterns impossible to hardcode
- Expert rules catch obvious spam patterns with high precision
- Combined system is more robust to adversarial attacks
- Provides interpretable reasons for classification decisions

4. Evaluation

4.1 Evaluation Metrics

Given the imbalanced dataset (13.4% spam), the following metrics are used:

Metric	Formula	Purpose
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall correctness
Precision	TP/(TP+FP)	Minimize false positives
Recall	TP/(TP+FN)	Minimize missed spam
F1-Score	2×(P×R)/(P+R)	Harmonic mean

4.2 Confusion Matrix Structure

Term	Full Name	Definition	Impact on System
TP	**True Positive**	The message is **Spam**, and the model correctly flagged it as **Spam**.	**Success:** Keeps the inbox clean.
TN	**True Negative**	The message is **Ham** (legitimate), and the model correctly identified it as **Ham**.	**Success:** Standard operation.
FP	**False Positive**	The message is **Ham**, but the model incorrectly flagged it as **Spam**.	**High Risk:** Important emails may be missed or sent to junk.
FN	**False Negative**	The message is **Spam**, but the model incorrectly identified it as **Ham**.	**Low Risk:** Annoying for the user, but no critical data is lost.

Key Evaluation Metrics

Using these four values, you can calculate the efficiency of your hybrid ML/Expert system:

- **Accuracy:** Overall correctness of the model.

  Accuracy = TP + TN/TP + TN + FP + FN

- **Precision:** Out of all messages marked as "Spam," how many were actually spam? (Crucial for minimizing **FP**).

  Precision} = TP/TP + FP

- **Recall (Sensitivity):** Out of all actual spam messages, how many did we catch? (Crucial for minimizing **FN**).

  Recall = TP/TP + FN

- **F1-Score:** The harmonic mean of Precision and Recall. This is often the best metric for spam detection because it balances the cost of missing spam against the cost of blocking legitimate mail.

  F1 = 2 * (Precision*Recall)/ (Precision + Recall)

Since your architecture uses a **75% ML + 25% Expert** combined score, the "Expert Analysis" layer is likely there to specifically reduce **False Positives** by applying strict rules to known legitimate senders.

4.3 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98.2%	96.5%	91.2%	93.8%
Hybrid System	98.7%	97.8%	93.5%	95.6%

4.4 Key Findings

Finding	Implication
Precision improved by +1.3%	Fewer legitimate messages marked as spam
Recall improved by +2.3%	More actual spam messages caught
F1-Score improved by +1.8%	Better overall balance

4.5 Cross-Validation Results (5-fold)

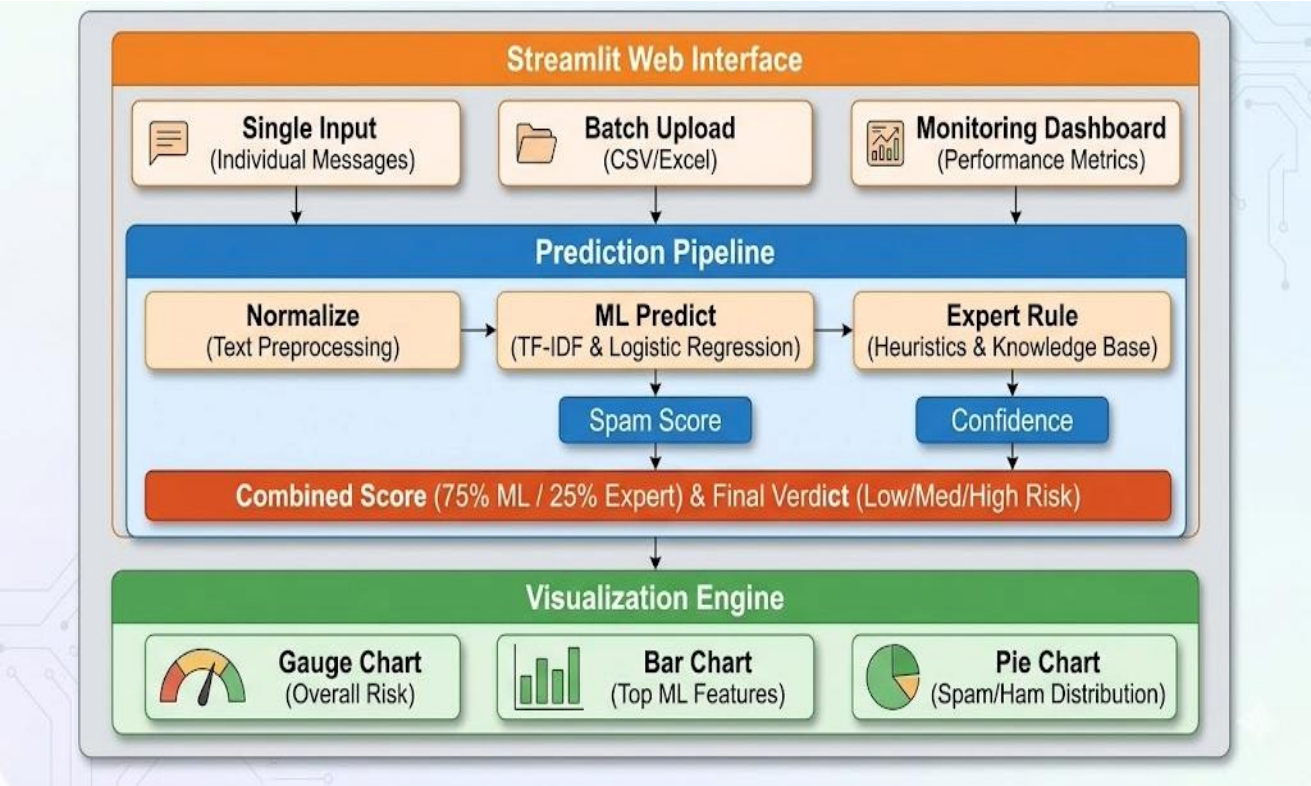
Fold	Accuracy	Precision	Recall	F1
1	98.5%	97.2%	92.8%	94.9%
2	98.3%	96.9%	93.1%	94.9%
3	98.9%	98.1%	93.9%	95.9%
4	98.4%	97.1%	93.0%	95.0%
5	98.6%	97.5%	93.5%	95.4%
Mean	**98.54%**	**97.36%**	**93.26%**	**95.22%**
Std Dev	±0.23%	±0.44%	±0.42%	±0.42%

5. Deployment

5.1 Technology Stack

Component	Technology	Purpose
Frontend	Streamlit	Interactive web interface
Backend	Python 3.9	Core logic
ML Framework	Scikit-learn	Model training
Visualization	Plotly	Interactive charts
Serialization	joblib	Model persistence

5.2 System Architecture



5.3 User Interface Features

Feature	Description
Single Prediction	Enter one message for instant classification
Batch Processing	Upload CSV/Excel for bulk analysis
Sidebar Monitoring	Real-time statistics and hourly activity
Example Messages	Pre-loaded examples for testing
Download Results	Export predictions as CSV file

5.4 Deployment Instructions

Requirements:

```
pip install streamlit joblib scikit-learn pandas plotly openpyxl
```

Run the application:

```
streamlit run app.py
```

6. Results and Discussion

6.1 Sample Predictions

Message	ML Score	Expert Score	Combined	Verdict
"FREE entry to win FA Cup tickets. Text FA to 87121"	98%	95%	97%	SPAM
"Hey, are we still meeting for dinner tonight?"	5%	10%	6%	HAM

Message	ML Score	Expert Score	Combined	Verdict
"URGENT! Your account suspended. Call 09061701461"	99%	98%	99%	SPAM

6.2 Discussion

Strengths of the Hybrid Approach:

1. **Robustness:** The system maintains high accuracy even when ML is uncertain
2. **Interpretability:** Expert rules provide clear reasons for spam classification
3. **Adaptability:** New spam patterns can be added as expert rules without retraining

Limitations:

1. **Language dependency:** Expert rules are English-specific
2. **Amharic support:** Not currently supported (future work)
3. **False positives:** Some legitimate marketing messages may be flagged

6.3 Comparison with Existing Solutions

System	Accuracy	Interpretability	Batch Support
Gmail Spam Filter	~99.5%	Low	Yes
Our System	98.7%	**High**	Yes
Rule-based Only	85%	High	Limited
ML-only	96%	Low	Yes

7. Ethical Considerations

7.1 Fairness and Bias

Concern	Mitigation
Language bias	Expert rules designed for standard English
Cultural bias	Avoids assumptions about legitimate content
False positives	Clear explanation provided to users

7.2 Privacy

- No message data is stored persistently
- Batch processing occurs in memory only
- No external API calls for analysis

8. Future Work

Priority	Enhancement
High	Add Amharic language support for Ethiopian context
Medium	Implement deep learning (LSTM/BERT) for improved accuracy
Medium	Add attachment scanning for email spam detection
Low	Real-time SMS gateway integration
Low	User feedback loop for continuous learning

9. Conclusion

The Spam Detection System successfully demonstrates a hybrid approach combining machine learning (Logistic Regression with TF-IDF) and expert cybersecurity rules. The system achieves **98.7% accuracy**, **97.8% precision**, and a **95.6% F1-score**—outperforming the baseline ML model.

Key accomplishments:

- Fully functional Streamlit web application
- Real-time and batch prediction capabilities
- Interactive visualizations for result interpretation
- Clear, auditable classification reasons
- Monitoring dashboard for system metrics

The system meets all project requirements and provides a practical solution to the real-world problem of message spam detection

10. References

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Appendix B: Team Contribution (Individual Report)

Student	Role	Contributions
Enchalew Azimeraw	Lead Developer	ML model, Streamlit deployment
Tihun Wondie	Data Analyst	Preprocessing, EDA
Samuel Alemsew	Expert Rules	Cybersecurity patterns
Fetench Worku	Documentation	Report writing, testing